

ANIRUDH BHATTACHARYA

Physical Design Application Engineer | Fusion Compiler / ICC2 | PnR
Hyderabad, Telangana | anirudhbhattaacharya.github.io/portfolio
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Education

Jaypee University of Engineering and Technology

2019 - 2023

Bachelor of Technology in Electronics and Communications Engineering | CGPA: 7.4

Guna, Madhya Pradesh

Relevant Coursework

- VLSI Design • Digital Hardware Design • Digital Circuit Design • Analogue Electronics
- Semiconductor Devices • Static Timing Analysis • Physical Design Flow • CMOS Fundamentals

Experience

Synopsys Inc.

Apr 2025 - Present

Physical Design Application Engineer

Hyderabad, Telangana

- Debug live Fusion Compiler and ICC2 PnR databases for product teams. Same breaks a PD team hits: floorplan, placement, legalize, CTS / MSCTS, route, MCMM STA, UPF, IR / EM, Calibre DRC / LVS. Nodes: TSMC N5A, 40ULP, 180G and Intel 18A / IFS18A.
- Isolated a live Qualcomm PnR fail across placement, CTS and route. Shipped GSAE testcase gsaetc_fc_command_trace from sh_command_log_file and log_trace. Reused by PD / AE teams. Excellence Award, March 2026.
- Closed 100+ FC / ICC2 cases. Accounts include Qualcomm, Samsung, Microsoft, Sony, ST, Marvell, Toshiba, Nuvoton Japan, Arm, NXP, MediaTek. Wrote 30+ SolvNetPlus articles.
- Queue: compile_fusion / flatten, clock_opt / MSCTS crash, create_clock_skew_group vs set_clock_balance_points, CCD / useful skew, NDR on leaf clocks, poly_rule on N5A, magnet placement, legalizer variant swap, IP-edge congestion, hyper_route_opt, via ladder, off-grid M2, UPF always-on split fanout, PVT mismatch, toggle-rate power, Calibre LVS.
- Tcl: ECO displacement CSV after set_attribute eco_change_status = create_cell and place_eco_cells -eco_changed_cells; cells-outside-core with boundary + ray cast; legalizer CMD-005 isolation. Customer-side debug. Do not own a product tapeout.

Synopsys Inc.

Apr 2024 - Mar 2025

Physical Design Intern

Hyderabad / Bhubaneswar

- Fusion Compiler RTL-to-GDS training: floorplan, PnR, CTS, STA, PDN, DRC / LVS. Converted to Physical Design Application Engineer in April 2025.
- Ran DSO.ai with Fusion Compiler on SLM PVT Monitor IP at Intel 18A / IFS18A (process detector, voltage monitor, temperature sensor). Timing, power and block-level QoR.

VLSI Expert Private Limited

May 2022 - Feb 2023

VLSI Trainee, Physical Design

Noida, Uttar Pradesh

- RISC-V picorv32a from Verilog through floorplan, PDN, CTS, PnR, STA and signoff setup in ICC2 and PrimeTime. Training block, not a product tapeout.
- SDC and max / min parasitic corners for MCMM STA.

Selected Work

Work	Node / tool	What I did	Result
Qualcomm PnR GSAE testcase	FC, Tcl	Command-trace of a live PnR fail	Reused; Award Mar 2026
Legalizer CMD-005 testcase	FC / ICC2	Isolate prelegal_tcl_callback_command	Two GSAE-style cases
Customer FC / ICC2 queue	N5A, 40ULP, 18A	CTS, legalize, route, UPF, PV debug	100+ cases, 30+ articles
SLM PVT Monitor IP	Intel 18A / IFS18A	DSO.ai + Fusion Compiler QoR	Converted to AE

SolvNetPlus (published titles from the queue)

CTS balancing and clock skew group pins. CCD prepone / postpone limits. IP-edge congestion. DRC convergence in compile_fusion. Antenna and diode insert. Crosstalk on reg2out. Unconstrained ICG paths, DC vs PrimeTime. 24 published titles. Bodies stay off public pages.

Technical Skills

Languages: Tcl, Python, Perl, Make, Verilog, Linux

CAD / EDA Tools: Fusion Compiler, ICC2, PrimeTime, Design Compiler, DSO.ai, Calibre

Physical Design: Floorplan, Placement, Legalize, CTS / MSCTS, PnR, PDN, STA / MCMM, UPF, IR / EM,

DRC / LVS / Antenna, SDC, useful skew / CCD, TSMC N5A / 40ULP / 180G, Intel 18A / IFS18A

Leadership / Extracurricular

IETE Students Forum — Volunteer

Sep 2019 - Aug 2021

VR AR MR Club — Coordinator

Sep 2019 - Oct 2020